CS315 · ALGORITHMS ANALYSIS & DESIGN

Practice 3 From the 461 Final

أسئلة فاينل 461 اللي داخلة في الميد فقط

محارب

01 قبل ما تبدأ

هذي أسئلة مأخوذة من فاينل 461 (26/12/2024). أخذت منه بس الأسئلة اللي مواضيعها داخلة في الميد، وتركت أسئلة الـ graphs و Kruskal و Dijkstra و Bellman-Ford و Fibonacci heap لأنها بعد الميد.

ما عندي نموذج إجابة للفاينل، فالحلول من حلي. الحل في ملف منفصل.

الأسئلة 02

MCQ · FINAL Q1 Q1

3) If several elements are competing for the same bucket or position in the hash table, what is it called?

a. Diffusion b. Replication c. Collision d. Duplication

4) Which of the following is not a technique to avoid a collision?

a. Make the hash function appear random

b. Use the chaining method

c. Use uniform hashing

d. Increasing hash table size

SHORT ANSWERS · FINAL Q2 · 8 MARKS

Q2

A. What are the three steps of Divide and conquer approach

B. Give two examples of Sorting algorithms using the Divide and conquer approach

C. If we have a B-tree in the order of 7 find:

1\ Max number of child = _____

2\ Max number of key = _____

3\ Min number of key = _____

ANALYSIS · FINAL Q3 C

Q3

Analyse (in details) the following algorithm Theoretically

```
Algorithm F1(a, n)
1. w ← 0
2. c ← 1
3. while w < n² do
4. w ← w + 1
5. c ← c * a
6. return c
```

B-TREE · FINAL Q4 C

Q4

Assume you have an empty B-tree of degree 3.

Draw (in details) the insert operation for the following elements: 8, 9, 10, 11, 15, 20.